

Edge-AI for VRU Behavior Prediction Using GPU-Accelerated Immersive Simulation in a CAVE Environment

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Abstract

This project targets the *Robotics and Edge AI* theme of NVIDIA’s Academic Grant Program (July–September CFP). We propose a closed-loop framework that combines our CAVE VR environment with NVIDIA’s edge and robotics stacks (Holoscan, Isaac, Newton) to learn and deploy vulnerable road user (VRU) behavior models. We will capture full-body motion and scene context in a multi-projector CAVE, train spatio-temporal models on cloud A100 GPUs, and deploy quantized, TensorRT-optimized models on Jetson AGX Orin for real-time inference (target < 20 ms end-to-end). The system integrates Unity-based scenarios via the Omniverse Unity Connector and synchronizes simulation physics (PhysX) with perception and prediction. Deliverables include: (i) a reproducible framework blueprint mapping RTX PRO 6000 (visualization), A100 (training), and Jetson (edge inference); (ii) an anonymized VRU dataset with licensing and documentation; (iii) open-source training/inference code and Holoscan graphs; (iv) real-time benchmarks on Jetson; and (v) a publication.

Keywords: Edge AI; CAVE VR; Isaac; Keypoint Skeleton Models; Intention Prediction

NVIDIA Platforms

Software: CUDA; PyTorch; TensorRT; NVIDIA Holoscan SDK (for low-latency sensor/graph pipelines); Isaac (Sim/Lab) and Newton (for physically grounded agent behaviors); Omniverse with Unity Connector (asset/simulation interoperability); DeepStream for video IO where applicable; Unity (content authoring).

Hardware: RTX PRO 6000 for multi-channel CAVE rendering and visualization; A100 GPUs (cloud) for model training; Jetson AGX Orin for low-latency edge inference.

Dataset and Model

Number of TBs: 2–4 (storage for raw captures, processed keypoints, and renderings).

Data type: Time-aligned 3D keypoint skeletons (60–120 Hz), synchronized scene state (semantic context, traffic signals, curb/lanes)

Data: Fully in-house with consented participants; protocol pre-registered; GDPR-compliant.

Confidential or personal data: No direct identifiers; bodies represented as keypoints; video (if any) anonymized by defacing; cloud compute limited to anonymized artifacts.

Approximate number of parameters: 100M–300M (teacher/backbone); distilled/edge variants $\leq$ 50M with quantization (INT8) for Jetson; scalable to 1B for ablations on cloud only.

Geographic location: Munich, Germany.

Introduction

We introduce a GPU-accelerated CAVE framework to study and predict VRU behavior under realistic, controllable traffic stimuli. **Novelty:** (1) a human-in-the-loop, immersive data collection setup producing high-fidelity keypoint and context labels; (2) a unified stack from

capture → cloud training → edge deployment with Holoscan/Jetson; (3) tight coupling of perception/prediction with physically grounded simulation (Isaac/Newton) for closed-loop evaluation; and (4) an open, reproducible blueprint for smart-mobility research labs. **Socio-economic impact:** improved safety analytics for pedestrians, cyclists, and e-scooter riders; transferable edge inference for smart intersections and micromobility fleets; reduced prototyping cost by sharing open datasets and code.

Methods

Capture and preprocessing (RTX PRO 6000 node): Multi-camera/keypoint extraction (Unity-based mocap and/or RGB-D), synchronization, and scene-state logging (signals, lanes, curbs). Operators are orchestrated as Holoscan graphs to enable deterministic latency and modularity.

Training (cloud A100): Spatio-temporal backbones (stacked GRUs and GNNs with temporal attention) trained on anonymized sequences. Augment with domain randomization and simulated edge noise. Mixed precision (AMP), DDP across 4 A100s; checkpointing with NVIDIA resiliency extension.

Model compression & compilation: Structured pruning and knowledge distillation to edge-sized backbones ($\leq 50M$). Calibrate INT8 with representative sets; export ONNX; build TensorRT engines targeting Orin; evaluate DLA vs. GPU paths.

Edge inference (Jetson AGX Orin): Holoscan pipeline for ingestion → preproc → TensorRT inference → postproc; target < 20 ms e2e latency at 60 Hz skeletons (latency budget: preproc 3 ms, inference 12 ms, postproc 5 ms).

Closed-loop evaluation (Isaac/Newton): Predicted intentions/trajectories influence agent controllers in Isaac/Newton while humans interact in the CAVE. Safety envelopes (time-to-collision, gap acceptance) measured online.

Visualization stack (RTX PRO 6000): Two GPUs drive multi-projector CAVE rendering (Mosaic/Frame lock); in addition they are used for visualization/recording and Omniverse simulation. Unity ↔ Omniverse via the Unity Connector keeps assets/scenes consistent.

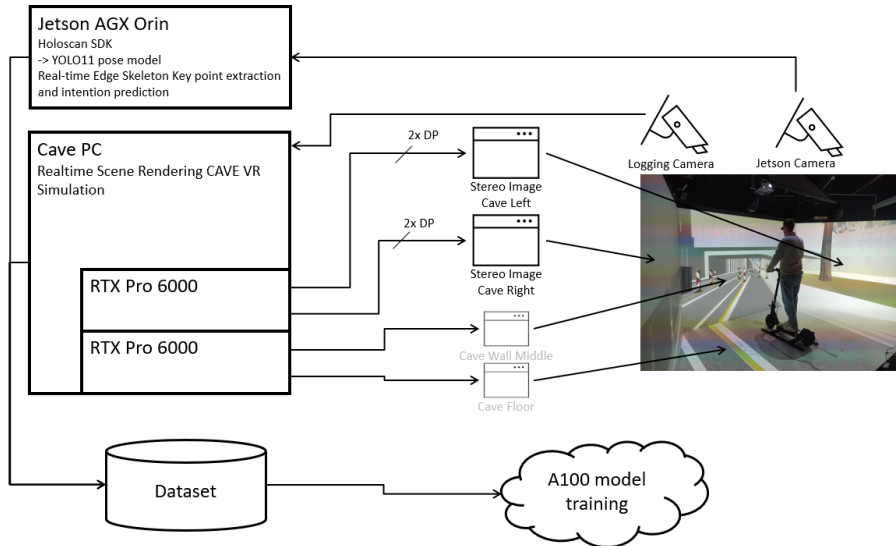


Figure 1: System blueprint and resource mapping: CAVE capture (RTX PRO 6000), cloud training (A100), edge inference (Jetson AGX Orin), and simulation coupling.

Expected Results

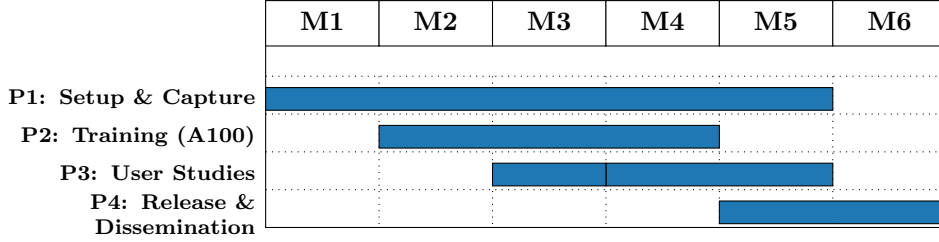
P1: CAVE setup; sensor/calibration; Unity↔Omniverse connector; pilot data captures. *(expected physical HW access in M3)*

P2: Baseline models; ablations; export to ONNX; TensorRT engine builds for Orin.

P3: Jetson deployment; real-time benchmarks; CAVE user studies, closed-loop evaluation.

P4: Dataset setup; open-source code release (GitHub, Apache-2.0); publication and demo video.

NVIDIA Cloud Use: Up to 10,000 GPU-hours on A100 (cloud) across months 2–5 (4 concurrent GPUs) for training, ablations, and quantization calibration. *(Availability expected M2)*



Projected Outcomes:

- One peer-reviewed publication.
- Public anonymized VRU dataset (CC BY 4.0), splits, and documentation (Hugging Face).
- Open-source training/inference code (Apache-2.0) and Holoscan graphs (GitHub).
- Framework blueprint and best-practice guide for CAVE→cloud→edge workflows.

Project Support Details

Hours in the cloud needed: 10,000

Number of concurrent GPUs needed: 4

Amount of cloud storage needed: 10 TB

Physical hardware needed: 2 × RTX PRO 6000 (Blackwell); 2 × Jetson AGX Orin

Rationale: A100 is requested as cloud resource only (no on-prem). Two RTX PRO 6000: dedicated to time-critical multi-projector CAVE rendering and for capture/visualization/Omniverse interop. Two Jetson AGX Orin: One in the Cave VR simulation lab and another one for optional field tests, depending on time constraints. Cloud hours (10k) reflect end-to-end training of 100–300M parameter models with ablations, plus distillation/quantization sweeps; this corresponds to approx. 4 months of effective use at 4 concurrent A100s with duty cycling.

Cloud Readiness

Operating System: Ubuntu Linux 22.04

Initial Steps: Secure ingest of anonymized data via `rsync/rc1one`; environment bootstrapping via Docker/containers (NVIDIA Container Toolkit); IaC scripts for reproducibility.

Experience:

- Data ingest: automated sync and integrity checks; GDPR-aware handling (keypoints only on cloud).
- Dependencies: proficient with apt/pip/conda and Docker/Singularity; CI for container builds.
- CUDA/NVIDIA SDKs: daily use; expanding to Holoscan, Omniverse Unity Connector, Isaac/Newton.
- Training/Serving: DDP on A100; ONNX export; TensorRT build; Jetson deployment with Holoscan graphs.
- Prior tech: prior prototypes relied on generic PyTorch and Unity-only pipelines; this project adopts NVIDIA Holoscan/Omniverse/Isaac/Newton to deliver real-time edge deployment and sim-to-real consistency.

Appendix A – Previous Collaboration with NVIDIA

N.A.

Appendix B – Project Collaborator CV

Dr.-Ing. Mathias Pechinger

Lead Researcher for VR simulation & AI for VRU behavior; cooperative PhD (TH Augsburg — TU München — Siemens); based in Germany.

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Focus

AI-assisted VR/AR traffic simulation for VRUs: real-time CAVE/HMD rendering, HD-map integration (OpenDRIVE/CityGML), deep learning for trajectory/intention prediction and human-state assessment.

Highlights

- GPU-accelerated multi-wall CAVE stack (Unity, NVIDIA GPUs; synced agents)
- Bike/e-scooter VR simulator with SUMO co-sim via OpenDRIVE; *Sumonity*
- HD-map fusion: RoadRunner with LOD3 CityGML for VR/AR studies
- Human-factors protocols (stress, rideability; e-scooters/bikes; geo-fencing)
- Tooling: Docker, Git CI; ROS; Expo/React Native

Skills & Education

Python, PyTorch; C/C++; Unity (C#); OpenDRIVE, SUMO; RoadRunner, CityGML; VR/CAVE & HMD workflows; Docker, GitLab CI; Linux/Ubuntu; Firebase, REST, Socket.IO; Raspberry Pi, Zigbee/MQTT.

Education: Dr.-Ing. (cooperative: TH Augsburg; TU München; Siemens); M.Sc. Applied CS (TH Augsburg); B.Eng. Electrical Engineering (TH Augsburg)

Selected Publications

- Zheng, T., Zhang, Y., **Pechinger, M.**, Lindner, J., & Bogenberger, K. (2025). Cyclist Maneuver Prediction at Unsignalized Intersections using a VR-based Bike Simulator. IEEE MT-ITS 2025; in press (publ. 1 Oct 2025).
- **Pechinger, M.**, & Lindner, J. (2024). Sumonity: Bridging SUMO and Unity... SUMO Conf. Proc., 5, 163–177. DOI: 10.52825/scp.v5i.1115.
- Lengerer, D., **Pechinger, M.**, & Markgraf, C. (2023). Transfer-Learning of a LiDAR Detection Algorithm... IEEE ITSC, pp. 3736–3741.
- **Pechinger, M.**, Lindner, J., Kolb, M., & Bogenberger, K. (2025). E-Scooter Safety Assessment using Geofencing at Urban Intersections: A Virtual Reality Study. IEEE MT-ITS 2025; in press (publ. 1 Oct 2025).
- **Pechinger, M.**, Schröder, G., Bogenberger, K., & Markgraf, C. (2023). Roadside Infrastructure Support for Urban Automated Driving. IEEE T-ITS. DOI: 10.1109/TITS.2023.3277138.

Appendix C – Citations

N.A.